

1. PROBLEM STATEMENT

Modern transactional systems (e.g., retail point-of-sale, transit turnstiles) assume continuous, low-latency, synchronous cloud connectivity. In real-world physical environments—underground metro tunnels, rural agricultural hubs, basements, and disaster zones—cellular networks suffer dead zones and frequent partitions. When connectivity drops, synchronous cloud RPC fails, halting transactions.

2. WHAT INDRA IS

Project INDRA (Intent-driven Networked Replication Architecture) is local-first state and transaction replication infrastructure. It provides a deterministic synchronization substrate combining state-based Conflict-Free Replicated Data Types (CRDTs), Radix-16 Merkle DAGs, and Delay-Tolerant Network (DTN) store-and-forward custody using opaque/encrypted relay envelopes. Nodes capture, validate, and commit state locally, propagating authorized mutations asynchronously whenever transient physical or multi-hop connectivity appears.

3. WHAT WAS DEMONSTRATED (EMPIRICAL EVIDENCE)

- Offline Endpoint Operation:** Two endpoints remain offline (cellular and Wi-Fi disabled; air-gapped from WAN connectivity during the experiment) while an intermediary relay reconnects the transaction path.
- Local Intent Authorization:** The endpoint unlocks local credential/key material and produces an Ed25519-signed PaymentIntent containing an opaque simulated HSM ciphertext payload without contacting any cloud server.
- Opaque Relay Custody:** Intermediary nodes act as voluntary, unprivileged DTN data mules, forwarding opaque transit envelopes across local radio/network links without access to plaintext transaction contents.
- Simulated Settlement Recording:** The Mock NPCI switch verifies the envelope signature, validates a 5-minute TTL window, checks a 16-byte deduplication index, and records simulated settlement.
- Sub-15ms Local Persistence:** Local WAL commit / gate-path durability was measured below 15 ms in the proving-ground run (P50: 0.06 ms collector SQLite commit, 11.12 ms turnstile gate clearance).
- Adversarial Resilience:** All five tested attack scenarios were rejected under the defined test conditions: (1) amount tampering, (2) replay of settled intent, (3) confirmation forgery, (4) expired TTL, and (5) relay snooping.
- Dynamic Relay Failover:** When the primary relay was disconnected mid-transaction, routing automatically transferred to a secondary relay with zero fund loss and zero double-debit.

4. EVIDENCE MATRIX: CLAIM VS. DEMONSTRATION SCOPE

Architectural Claim	Exact Demonstrated Evidence	Boundary / Operating Condition
Offline Endpoint Operation	Physical phone run (S24+ and S25 Ultra)	Cellular and Wi-Fi disabled; air-gapped from WAN
Local Transaction Creation	Physical phone run (node.viviktam.payer)	Local credential unlock, Ed25519 digital signature
DTN Relay Semantics	Anubandha proving ground (anubandha/src/relay.rs)	Opaque envelope store-and-forward custody
BLE Physical Transport	Laboratory RF test suite (crates/indra-transport/src/ble)	512B sub-MTU slicing; segregated from Anubandha run
Local Network Transport	Automated socket test suite (lan_transport_tests.rs)	In-process framed channels & local TCP loopback
Authority Settlement	Mock NPCI switch (anubandha/src/mock_npci.rs)	In-memory simulated authority; in-process ledger
Production Banking Integration	Not demonstrated	Explicit non-claim; requires formal regulatory clearance

5. WHAT WAS NOT DEMONSTRATED (EXPLICIT NON-CLAIMS)

- No Live Banking Integration:** No live NPCI, UPI, core-banking, or RBI clearing infrastructure is involved.
- Simulated HSM:** The settlement/HSM component in this proving ground is simulated; no production PCI-PTS hardware HSM integration is claimed.
- No Unbounded Anonymous Consensus:** INDRA does not claim open-ended Byzantine consensus across anonymous untrusted networks; correctness relies on cryptographic authority signatures, nonces, and bilateral peering policies.

“Here is what we built. Tell us where this architecture fails.”